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Exp no:2

Developing and Deploying a Simple Chaincode

Aim

To develop, package, install, approve, commit, and execute a simple asset transfer chaincode in a Hyperledger Fabric test network using the JavaScript programming language

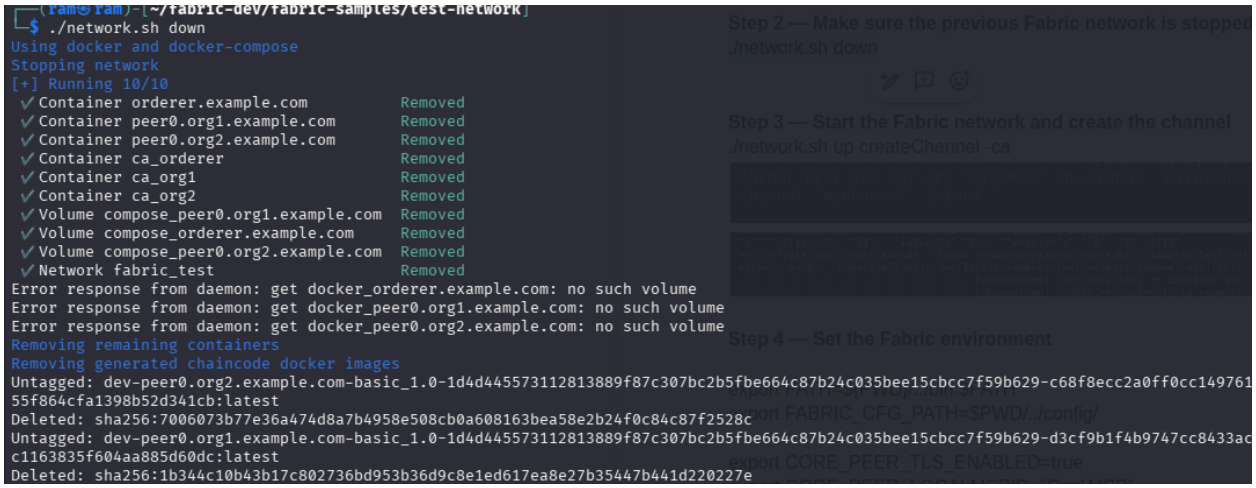
Procedure:

Step 1 — Go to the test network

cd ~/fabric-dev/fabric-samples/test-network

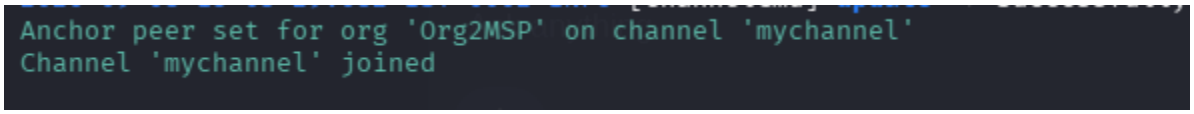
Step 2 — Make sure the previous Fabric network is stopped

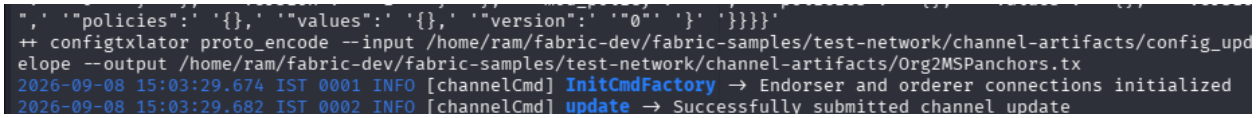
./network.sh down



Step 3 — Start the Fabric network and create the channel

./network.sh up createChannel -ca





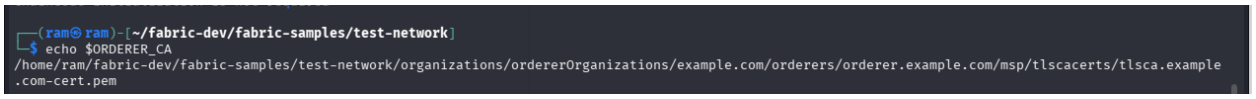
Step 4 — Set the Fabric environment

export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=$PWD/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051

export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem

echo $ORDERER_CA
It should print the path to the orderer's TLS CA certificate.



Step 5 — Deploy the JavaScript chaincode
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript

Step 6 — Verify that the chaincode is installed
peer lifecycle chaincode queryinstalled

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ peer lifecycle chaincode queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f59b629, Label: basic_1.0
```

Step 7 — Initialize the ledger
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"InitLedger","Args":[]}'

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"InitLedger","Args":[]}'
2026-09-08 15:12:05.394 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200
```

Step 8 — Query all assets
peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'

```
┌──(ram㉿ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":800,"Color":"Black","ID":"asset10","Owner":"David","Size":15},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
```

Step 9 - Create asset10
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \

```
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"CreateAsset","Args":["asset10","Black","15","Arun","800"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.c
om-cert.pem" \
--waitForEvent
```

Read it

```
peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset10"]}'
```

Transfer it to David

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"TransferAsset","Args":["asset10","David"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.c
om-cert.pem" \
--waitForEvent
```

Step - 10 verify

peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset10"]}'